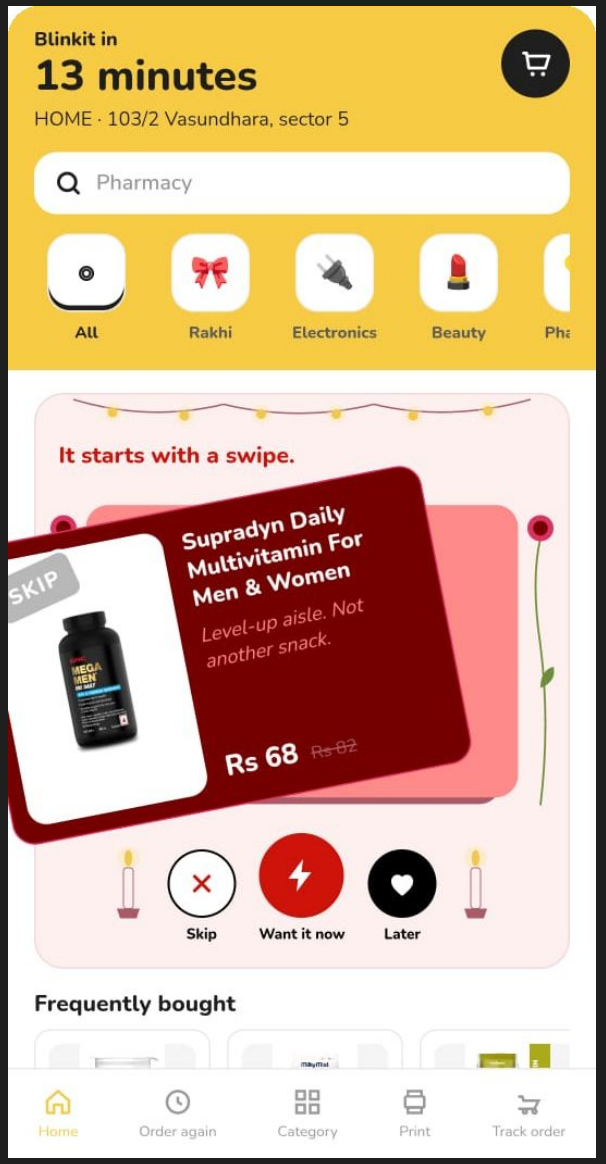
NL BLINKIT

# Getting Blinkit users out of the staples loop

I started with 156,219 Play Store reviews, narrowed them to one validated problem, and shipped a swipe-to-discover MVP against it.

**Growth goal:** increase the % of Monthly Active Customers who buy from at least one new category each month

*AI did the heavy lifting at every step: scraping by code, classifying and clustering by models, recommending by an LLM ladder.*



# One AI workflow turned 156,219 raw reviews into the 1% that actually matters

1. **Scrape.** 156,219 Blinkit Play Store reviews via google-play-scraper, Jan 2 to Jul 28, 2026.

2. **Clean & split.** Strip junk, keep Devanagari and emoji, split into single complaints, drop fragments. 84,111 units.

3. **Classify addressability.** Tag every unit app_ux / operational / pricing / praise. Keyword filter first, Groq llama-3.1-8b for unclear ones.

4. **Cluster.** MiniLM embeddings, kNN graph (k = 15), Louvain finds the natural communities.

5. **Name themes.** An LLM writes a short problem statement per community, plus its link to category exploration.

6. **Interrogate.** A RAG chatbot on the corpus. Every claim must cite real reviews; inferences get labelled.

**Why step 3 exists:** my first run surfaced only operational pain (order issues 4,873, weak support 4,452, late delivery 4,264). Real, but nothing an app growth PM can ship. So I added the gate, and re-ran.

**84,111 units, classified:**



**53,525** (63.6%) Praise / noise

**25,808** (30.7%) Operational

**3,936** (4.7%) Pricing & policy

**842** (1.0%) app_ux, addressable in the product

**Quality checks:** modularity 0.7519 (strong communities) · 12/15 themes stable across cohorts · citations mandatory in chat · stress-tested against 4 live interviews. Method adapted from Tabayashi (2026).

# Inside those 842 units, one theme sits exactly where category exploration happens

| # | app_ux theme (final run) | Units | % of 842 | What it really is |
|---|---|---|---|---|
| 1 | Payment method unavailability | **119** | 14.1% | Checkout plumbing |
| 2 | App glitches & bugs | **94** | 11.2% | App stability |
| 3 | Cart & order issues | **89** | 10.6% | Checkout plumbing |
| 4 | **Irrelevant product recommendations** | **85** | 10.1% | Browse / discover, the metric's home |
| 5 | Blinkit Wallet unavailability | **76** | 9.0% | Checkout plumbing |
| 6 | Easy navigation | **63** | 7.5% | Findability |

**I picked on metric relevance, not raw count.** Themes 1, 3 and 5 are checkout hygiene, and #2 is stability. Fixing any of them makes checkout smoother, but none of them makes a user try a new category. #4 is the only top-5 theme living at browse and discover, which is exactly where my growth metric lives.

**What is inside theme #4:** irrelevant and out-of-stock suggestions, and old searches resurfacing instead of new products. The cluster formed purely algorithmically (community 999, journey stage: browse_discover). The pipeline surfaced it; I did not seed it.

# I sat with four users from the segment. They confirmed the theme without being asked

**The segment:** metro quick-commerce natives in their 20s. Mobile-first, 2-3 orders a week, narrow repeat baskets of dairy, staples and snacks, price and fee sensitive. They are active every month but rarely expand categories, which makes them the exact denominator of my growth metric.

### P Prateek · 26 · M

Orders 3x a week, mostly evenings. His newest buy, a phone charger, came via the search bar. A tiny Boat speaker and near-expiry skincare burnt his trust, so now he checks options, then price, and wants to feel a product before trying a new category.

### A Akash · 28 · M

Protein-heavy basket: dairy and oats. Newest buys, a bed table and a weight scale, both via search. He never opens category tabs or suggestions. He once searched his exact product and found it sitting below the ads.

### J Janvi · 24 · F

Stocks up monthly when salary lands, then re-orders about twice a week. Compares prices across every app before anything costly. A stale paneer and an expired washing liquid taught her to check expiry dates at the door, every single delivery.

### B Bardhan · 25 · M

The explorer: tries something new almost weekly, mostly imported items. His newest buy, a raincoat, came via search because it never appeared in the category list. He wants new products shown, not his recently viewed.

**What all four told me, unprompted:** 4/4 called the recommendations bad · 4/4 said ads bury the product they actually want · 3/4 found their newest purchase via search only. And one thing the engine under-weighted: trust runs deeper than the reviews showed, so every recommendation must justify itself.

# Blinkit's surfaces optimise for what you already bought, so baskets never grow sideways

### What the ranking visibly depends on (observed from reviews and interviews, not internal knowledge)

**Recency.** Recently viewed items and old searches keep resurfacing. Bardhan put it simply: show me new products, not what I already saw.

**Popularity.** Bestseller rails repeat the same mass-market items for everyone.

**Sponsored placement.** Ads rank above the relevant result. Akash's exact product appeared below the ads, and all four hit some version of this.

### Why that blocks the new-category metric

Recency reinforces the repeat loop instead of breaking it. Habits run cue, routine, reward, and familiar picks save mental effort (YourStory, 2026). The app has to supply the break trigger, and today it supplies the opposite.

Ads occupy the very slots where discovery should happen.

There is no cross-category inference. A high-protein basket never gets shown protein bars, a scale, or paneer. The sideways connection simply does not exist.

**What users do instead, their workarounds:** they treat the search bar as the only discovery tool, they compare prices on rival apps before anything expensive, and they fall back to the local store for items the app prices or bundles badly. The exploration intent clearly exists. The app just gives it no path.

# Problem Framing Canvas

## 1 · What is the true problem?

Users stay locked in repeat staples because the discovery surfaces serve recency and ads, not relevant new categories. Exploration only happens when a user already knows what to search for.

## 2 · Who is facing it?

High-frequency, narrow-basket quick-commerce users: metro, in their 20s, mobile-first, 2-3 orders a week, price and fee sensitive. High MAC, low category expansion.

## 3 · How do we know it is real?

Quant: 85 of 842 addressable app complaints (10.1%) are about irrelevant recommendations, the only top-5 theme at browse and discover.
Qual: 4/4 interviews said recommendations are bad, 4/4 said ads bury discovery, 3/4 found new items via search only.

## 4 · What is the real value of solving it?

This is a discovery-fit problem, not a demand problem. Users already want to explore; one interviewee tries something new weekly. The app gives that intent no path. Fixing fit converts existing intent into new-category orders, instead of paying to create demand.

## 5 · Why solve it now?

**For users**

- Relevant new products with zero effort
- Discovery becomes play, not a chore
- Nudges land when price or stock actually fits

**For business**

- Directly moves the MAC new-category metric
- Wider baskets lift AOV and retention
- 71% expect personalisation and 76% get frustrated without it (McKinsey), a gap rivals will close first

# Walk with one user: from “never knew it existed” to a new-category order

*Take Akash. Protein-heavy basket, orders in the evening, never opens category tabs.*

## TODAY: THE STAPLES LOOP

Opens the app with milk and oats in mind → Searches the exact item, scrolls past the ads → Checks out the same staples as always → Recommendations show his recently viewed, again → Nothing new ever enters the basket

## WITH THE DISCOVERY DECK: THE LOOP BREAKS

Same start. The deck sits below search, never in the way of an urgent order → While the rider is on the way, he plays a few swipes → Cards speak his language: “Malai kam, protein zyada” ties Skyr to his own dairy basket → Even skips do work. Card by card, he now knows these products exist → A right swipe parks his curiosity in Saved, no commitment needed → The nudge lands at his usual evening order time, with the price dropped. He is ordering anyway, so it simply joins the basket

**Two quiet mechanisms do the selling.** Unconscious observation: every card seen is a product he now knows exists, even the ones he skips. And the natural clock: the nudge fires inside the window when he orders anyway, so trying something new costs zero extra effort and zero extra delivery fee.

# The metric splits into three levers, and the MVP pulls all three

*The logic in one breath: nobody buys a category they never see, nobody engages a card they do not trust, and everybody converts the moment the price objection dies.*

**% of MACs buying ≥ 1 new category / month**

**EXPOSURE**
Do they even SEE a relevant new-category item?

The ladder of inference starts from what I know: L0 basket facts, L1 needs and hard constraints like veg or budget. From there it climbs sideways: L2 same need in a new category, L3 same goal, L4 lifestyle halo. Groq llama-3.3-70b generates the candidates per persona, and a deterministic allocator serves the deck, so it is instant, predictable, and free per swipe.

**CONSIDERATION**
Do they TRUST it enough to engage?

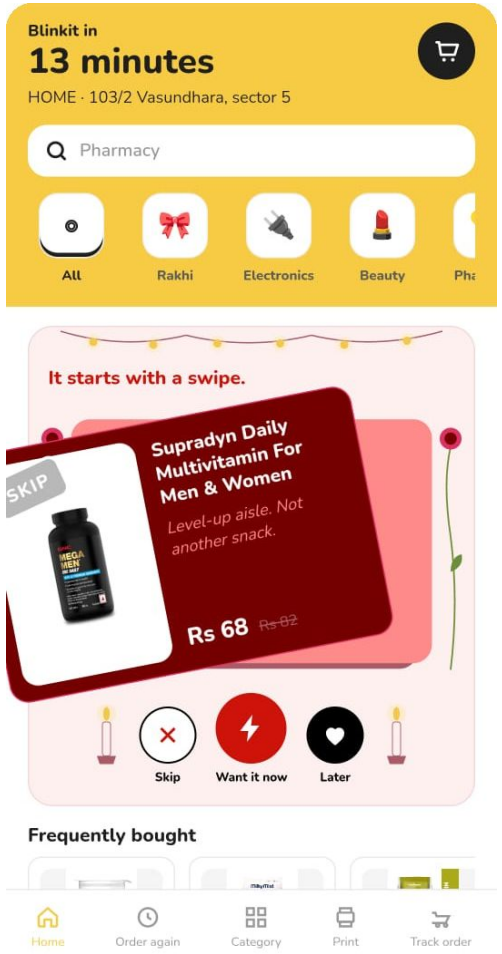
Every card must carry a one-line bridge back to real basket items. No honest bridge, no card. And the deck respects nerves: stage 0 serves 4 safe L2 picks and 1 L3, engagement unlocks bolder L4 leaps, and two dismissals in a row retreat to safe L2. Habits break through small safe steps, not big leaps.

**CONVERSION**
Does it reach the cart at the right moment?

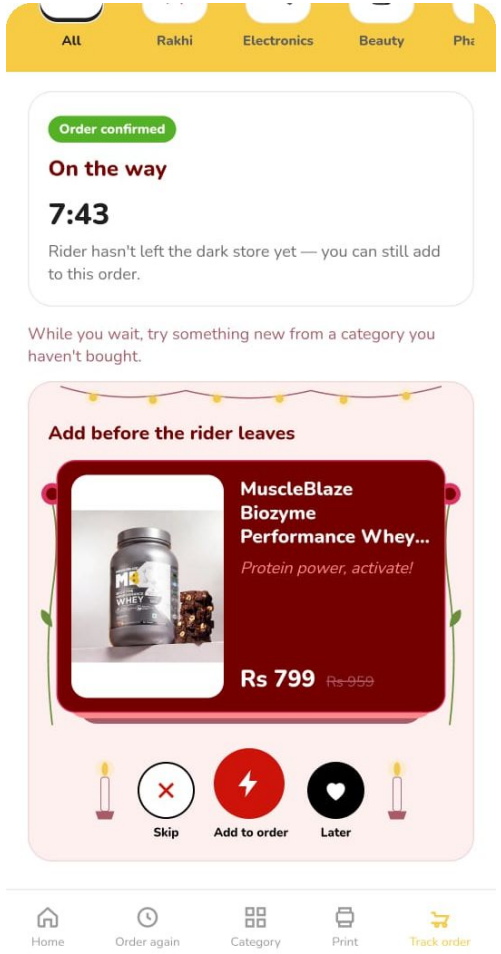
A right swipe parks the item in Saved. Then the app waits for the moment the excuse dies: a price drop, a free-delivery gap of ₹70 or less under the ₹99 threshold, or a restock. That is when the Match fires. An up swipe skips the wait and adds to cart or the live order immediately.

**What I measure:** swipe participation rate · right-swipe rate · saved to cart conversion · nudge CTR · new-category repeat rate. Each one rolls up into one of the three levers above.

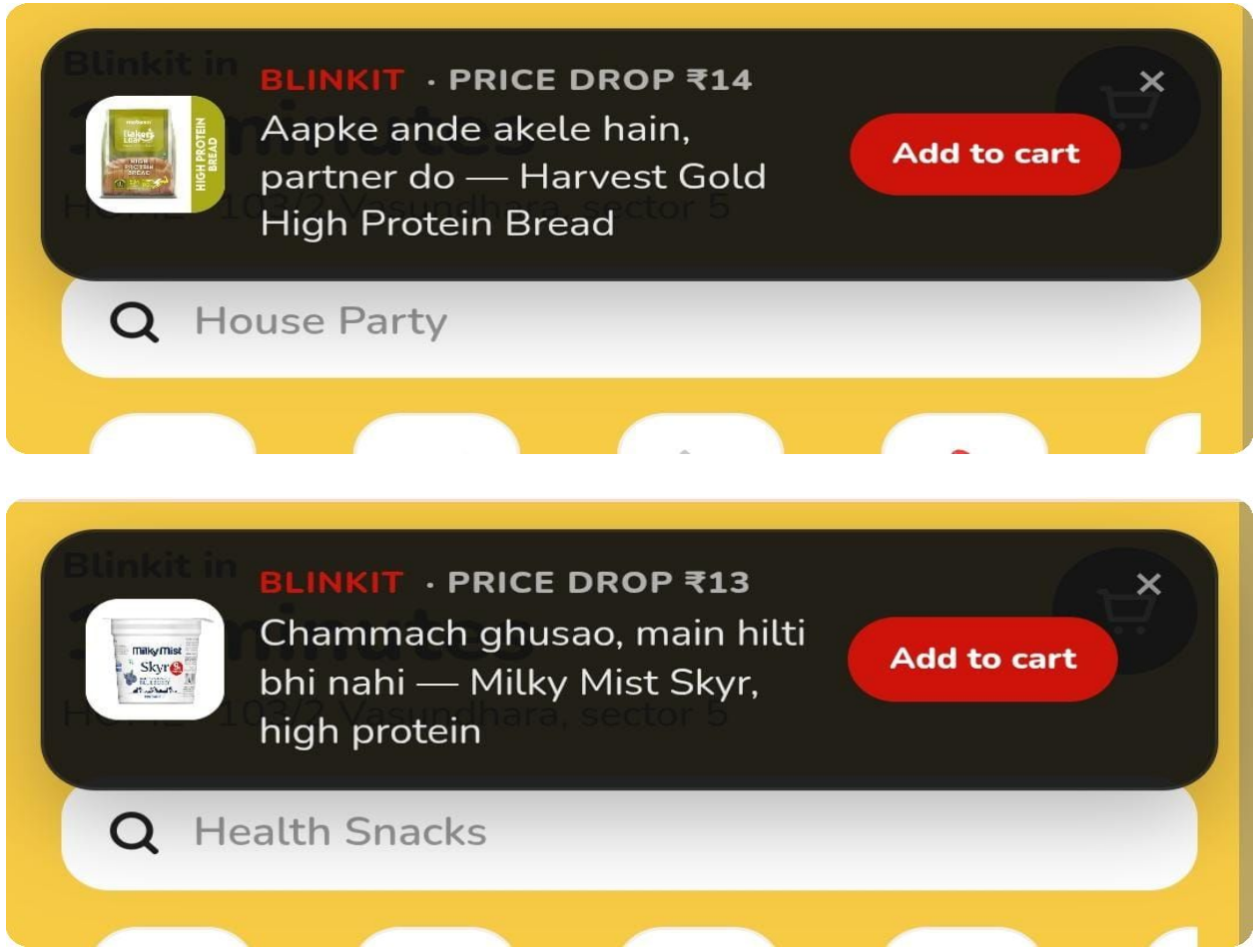
# The MVP, live: a cross-category engine wearing a Tinder skin




Home feed. The deck lives in Blinkit's own themed-banner slot, below search, so urgent orders are never blocked. Skip / Want it now / Later.




Order tracking. While the rider is on the way, it invites you to try a category you have not bought. An up swipe adds to the live order.




*A right swipe parks the item in Saved, and when the trigger fires, a Hinglish match nudge lands (price drop shown). Sample copy; the LLM generation path is built in.*

# Everything is live. Try it

### AI Discovery Engine

https://blinkit-ai-discovery-engine-91md.vercel.app/

### Deployed MVP

https://blinkit-iota-ruddy.vercel.app

### Primary research

4 moderated user interviews with the target segment: Prateek, Akash, Janvi, Bardhan

Scan to open the MVP on your phone. It is built like an app, so mobile s the best way to feel it.

### References


- McKinsey & Company (2021), Next in Personalization: 71% of consumers expect personalised interactions, and 76% get frustrated without them.
- Tabayashi, K. (2026), Extracting Insights from Customer Feedback Using LLMs and Graph Algorithms, Medium: the methodological basis for the embed, kNN, Louvain, LLM pipeline.
- YourStory (Jan 2026), The psychology behind why we buy the same thing again and again: habit loops and choice overload behind the staples default.
- Frontiers in Communication (2025): gamification in m-commerce raises engagement and reduces switching behaviour.


*Stack: google-play-scraper · MiniLM · Louvain · Groq (llama-3.1-8b classify & name · llama-3.3-70b recommend) · React · Railway + Vercel*